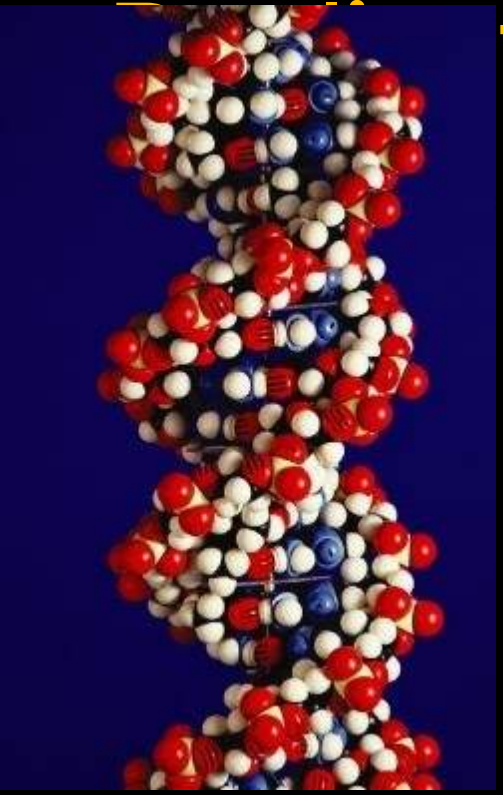
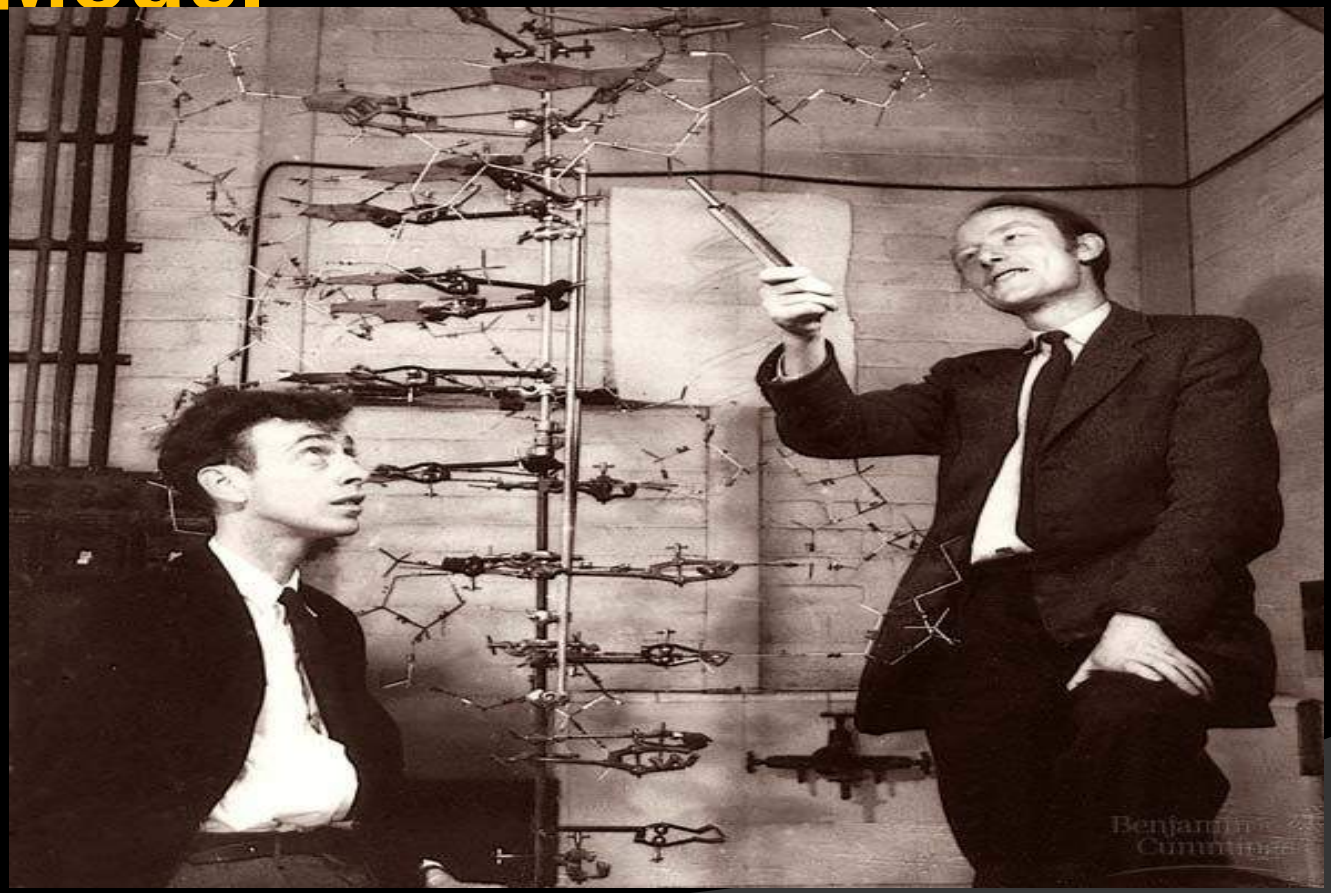
# DNA

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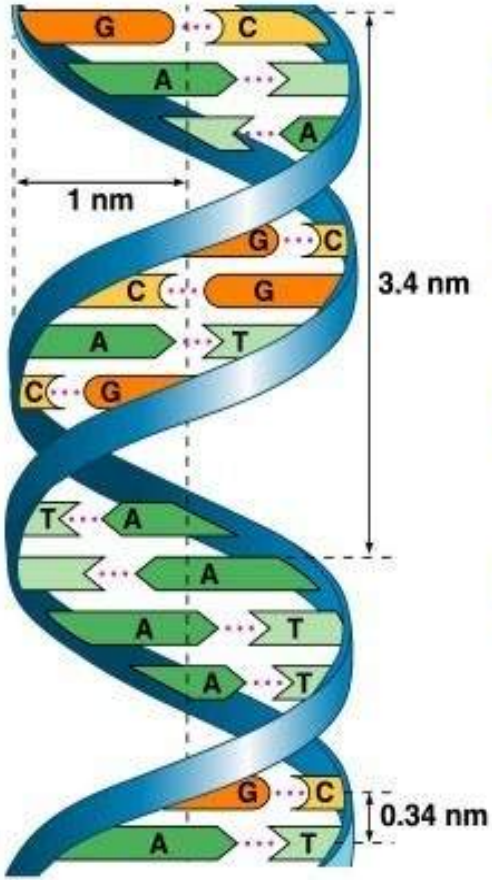




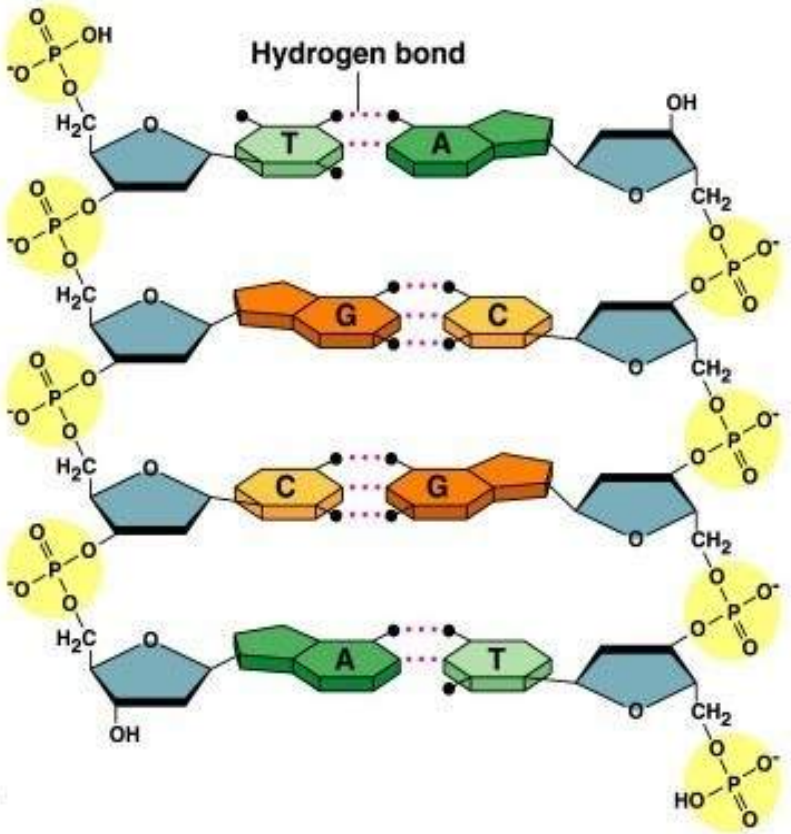
# DNA – DOUBLE HELICALSTRUCTURE WATSON and CRICK-Model



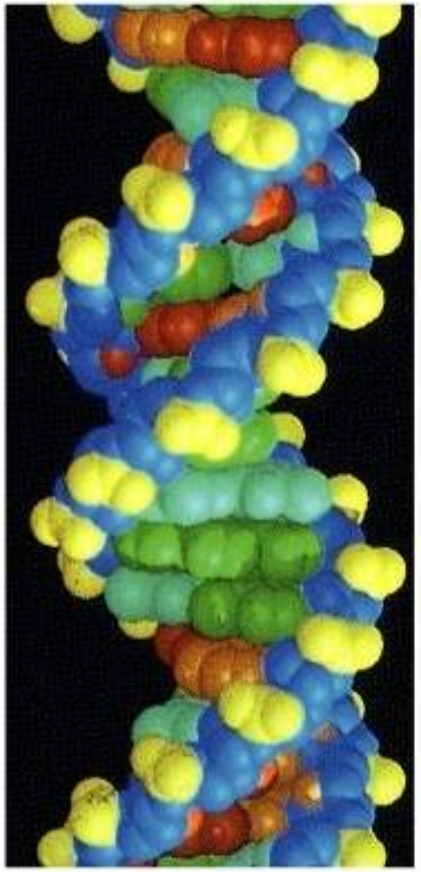
# DNA – DOUBLE HELICALSTRUCTURE




(a) Key features of DNA structure

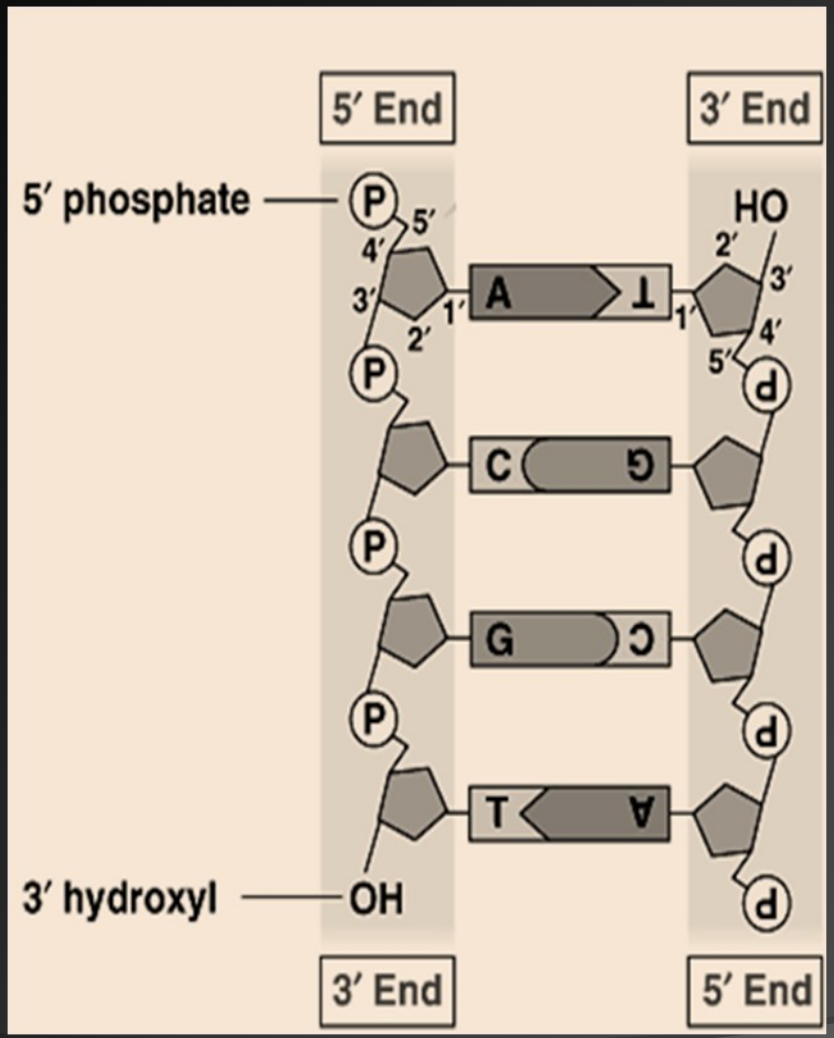



(b) Partial chemical structure



(c) Space-filling model

# Directionality of DNA




Nucleotides in DNA backbone are bonded together by phosphodiester linkage between 3′ & 5′ carbons.

DNA molecule has “direction.“

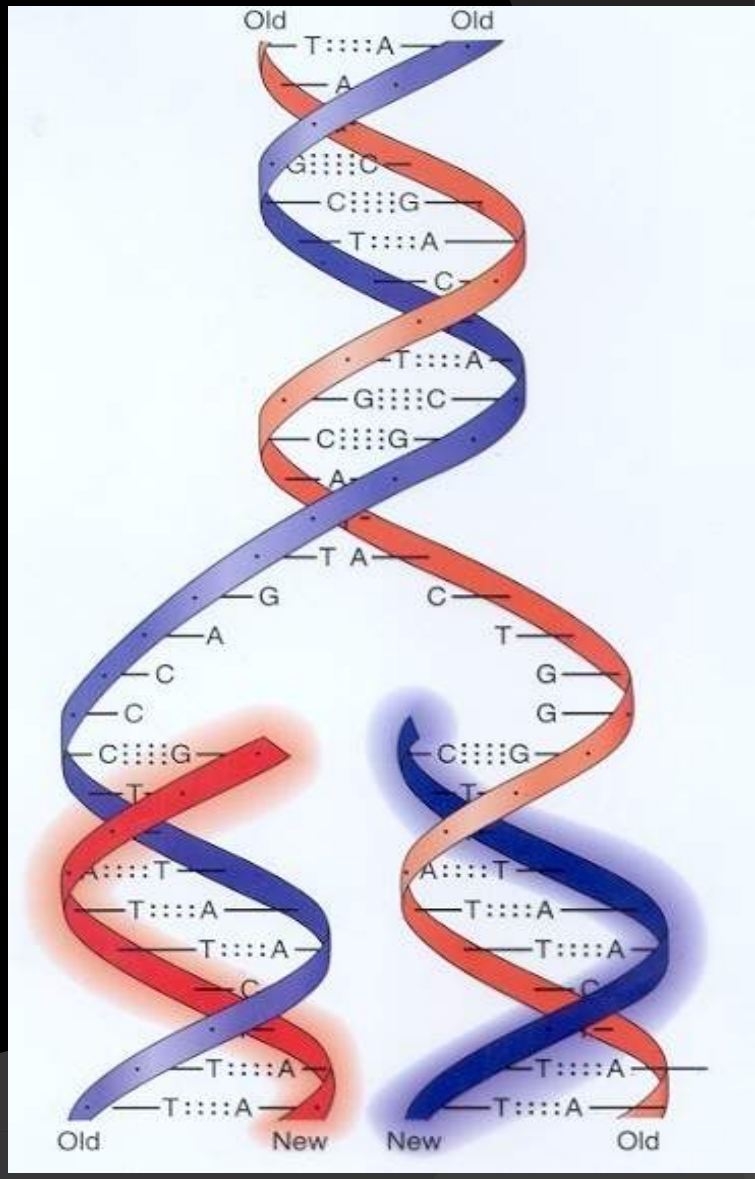
Complementary strands run in opposite directions.

# DNA Replication- Introduction

- Basis for inheritance
- Fundamental process occurring in all cells for copying DNA to transfer the genetic information to daughter cells
- Each cell must replicate its DNA before division.

# DNA Replication

- Semi conservative
- Parental strands are not degraded
- Base pairing allows each strand to serve as a template for a new strand
- New duplex is 1/2 parent template & 1/2 new DNA



# DNA Replication



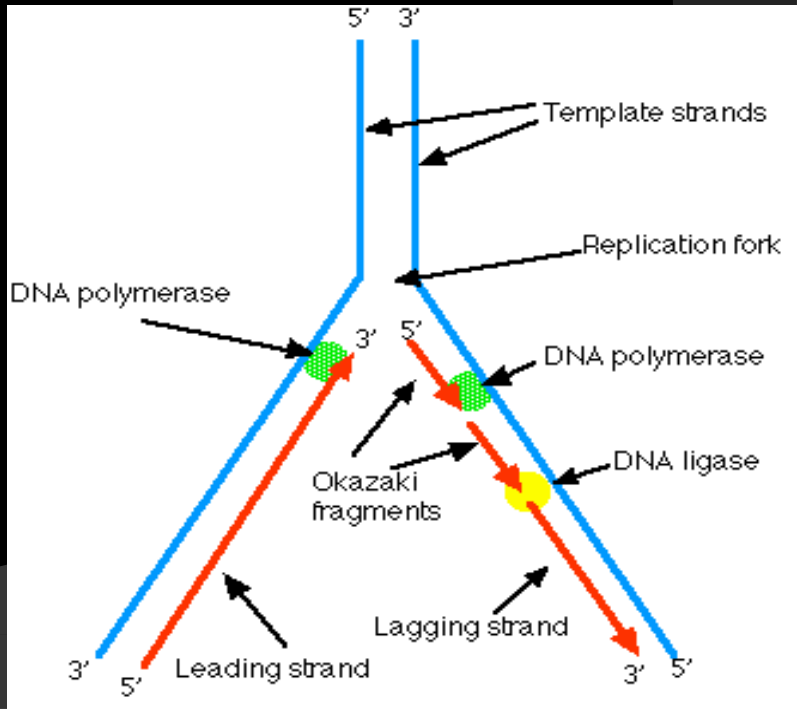

- Semi discontinuous
- Leading & Lagging strands

**Leading strand**

- **continuous synthesis**

**Lagging strand**

- **Okazaki fragments**
- **joined by ligases**



# DNA Replication

- Energy of Replication
- The nucleotides arrive as nucleoside triphosphates
  - DNA base, sugar with PPP
    - P-P-P = energy for bonding
  - DNA bases arrive with their own energy source for bonding
  - bonded by enzyme: DNA polymerase III

# DNA Replication

- Primer is needed
  - DNA polymerase can only add nucleotides to 3′ end of a growing DNA strand
    - need a “starter” nucleotide to make a bond
  - strand only grows 5′→3′
  - Template is read in the 3′-5′ direction while polymerization takes place in the 5′→3′ direction

# Primer

**RNA primer**

- Synthesized by Primase
- serves as a starter sequence for DNA polymerase III
- Only one RNA Primer-required for the leading strand
- RNA Primers for the lagging strand depend on the number of “OKAZAKI FRAGMENTS”
- RNA Primer has a free 3’OH group to which the first Nucleotide is bound.

# DNA Replication-Steps

- Identification of the origins of replication
- Unwinding (denaturation) of dsDNA to provide an ssDNA template
- Formation of the replication fork
- Initiation of DNA synthesis and elongation
- Primer removal and ligation of the newly synthesized DNA segments
- Reconstitution of chromatin structure

# Components of Replication

- DNA polymerases- Deoxynucleotide polymerization
- Helicase -Processive unwinding of
- DNA
- Topoisomerases Relieve torsional strain that results from helicase-induced unwinding
- RNA primase Initiates synthesis of RNA primers
- Single-strand binding proteins Prevent premature reannealing of dsDNA
- DNA ligase Seals the single strand nick between the nascent chain and Okazaki fragments on lagging strand

# Origin of Replication- Prokaryotes

- At the origin of replication **(ori),** there is an association of sequence-specific dsDNA-binding proteins with a series of direct repeat DNA sequences.
- In *E coli,* the oriC is bound by the protein dnaA.
- a complex is formed consisting of 150–250 bp of DNA and multimers of the DNA-binding protein. This leads to the local denaturation and unwinding of an adjacent A+T-rich region of DNA.

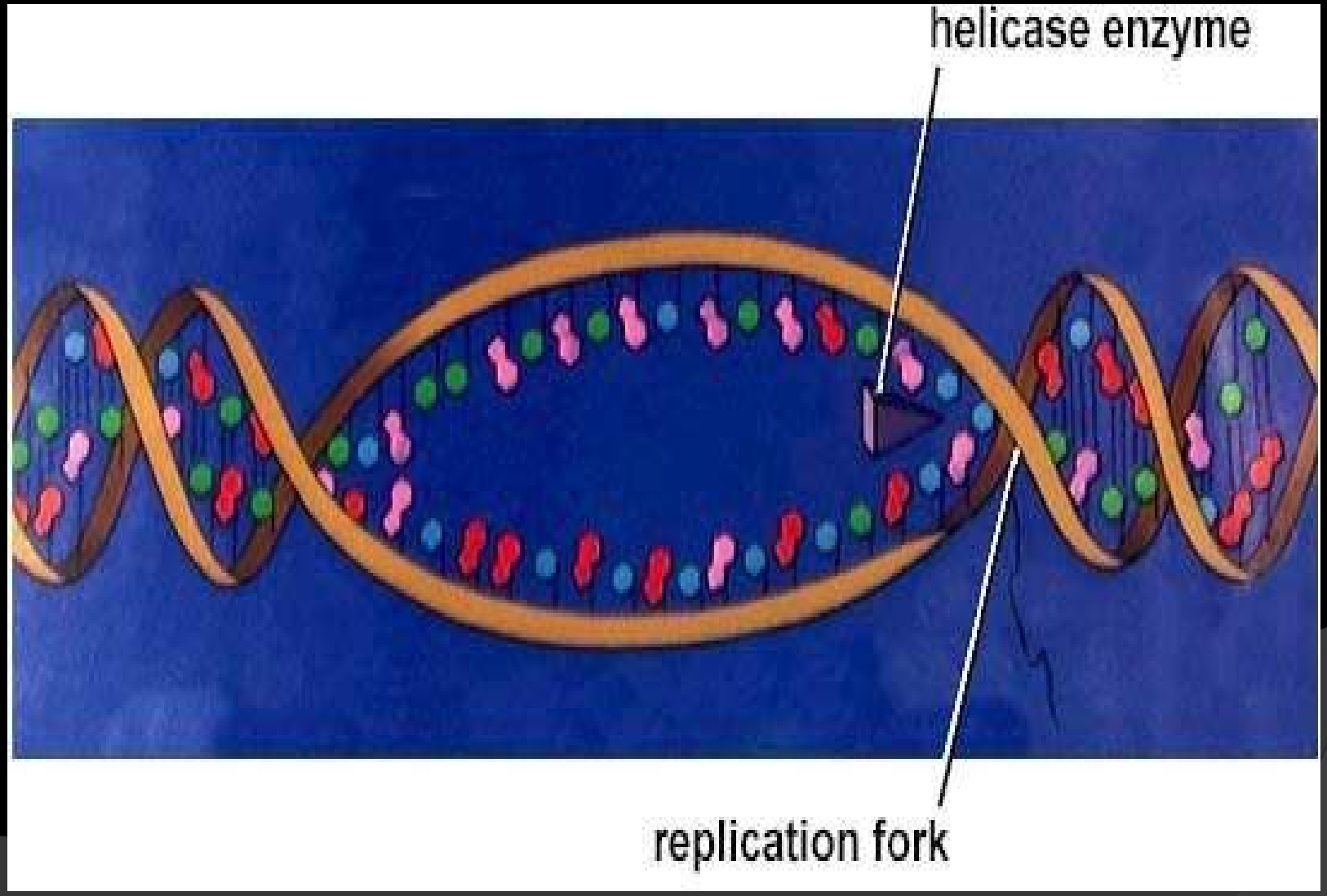
# Origin of Replication -Eukaryotes

- Functionally similar **autonomously replicating sequences (ARS) or replicators** have been identified in yeast cells.
- The ARS contains a somewhat degenerate 11-bp sequence called the **origin replication element (ORE).**
- The ORE binds a set of proteins, analogous to the dnaA protein of *E coli,* which is collectively called the **origin recognition complex (ORC).**
- The ORE is located adjacent to an approximately 80-bp A+T-rich sequence that is easy to unwind. This is called the DNA unwinding element (DUE**).**

# Unwinding of DNA

- The interaction of proteins with ori defines the start site of replication and provides a short region of ssDNA essential for initiation of synthesis of the nascent DNA strand.
- DNA Helicase allows for processive unwinding of DNA.
- Single-stranded DNA-binding proteins (SSBs) stabilize this complex.
- In cooperation with SSB, this leads to DNA unwinding and active replication.

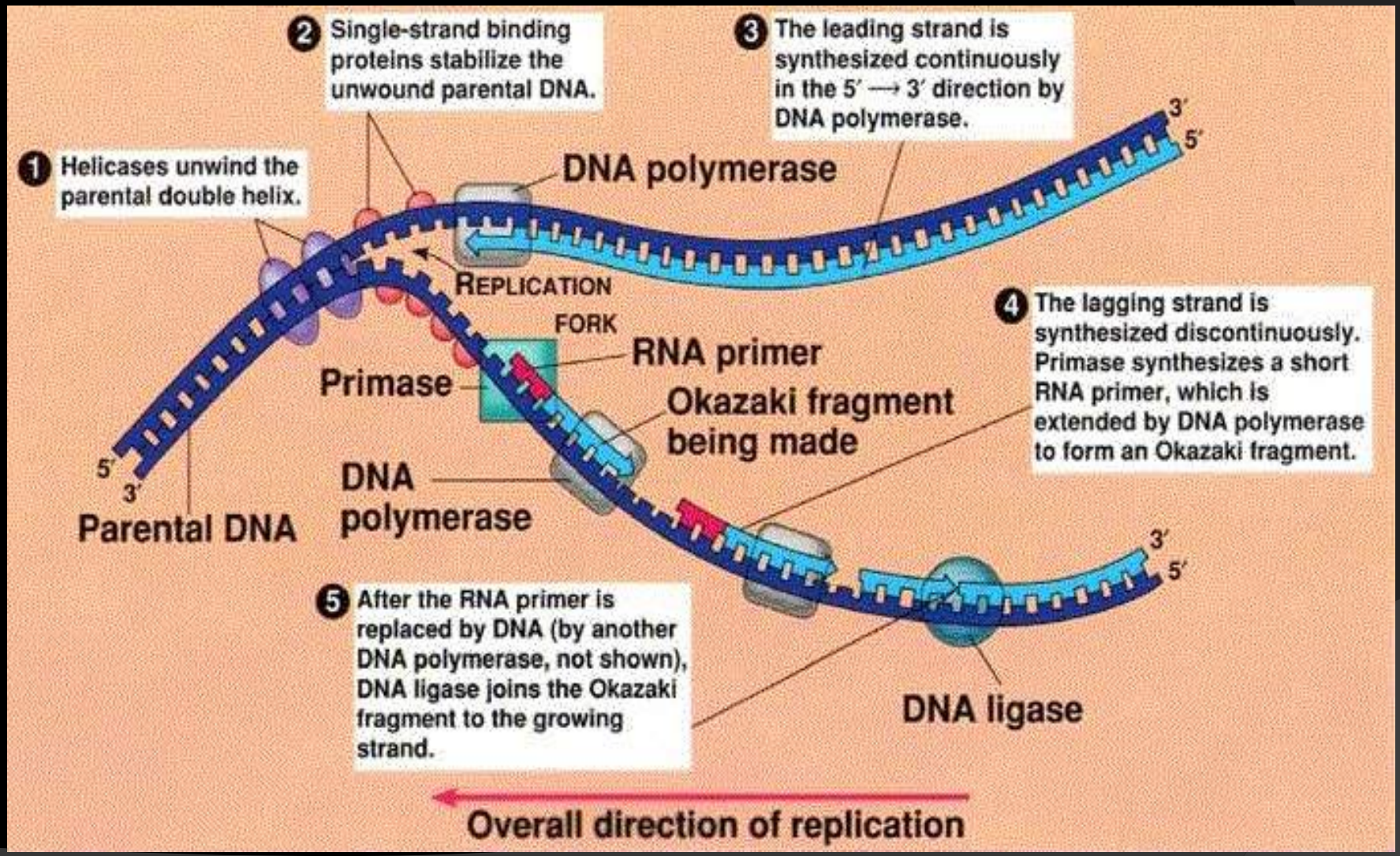
# Unwinding of DNA



# Formation of the Replication Fork

- The polymerase III holoenzyme binds to template DNA as part of a multiprotein complex
- DNA polymerases only synthesize DNA in the 5' to 3' direction,
- Because the DNA strands are antiparallel , the polymerase functions asymmetrically.
- On the leading (forward) strand, the DNA is synthesized continuously.
- On the lagging (retrograde) strand, the DNA is synthesized in short (1–5 kb)fragments, the so-called Okazaki fragments.

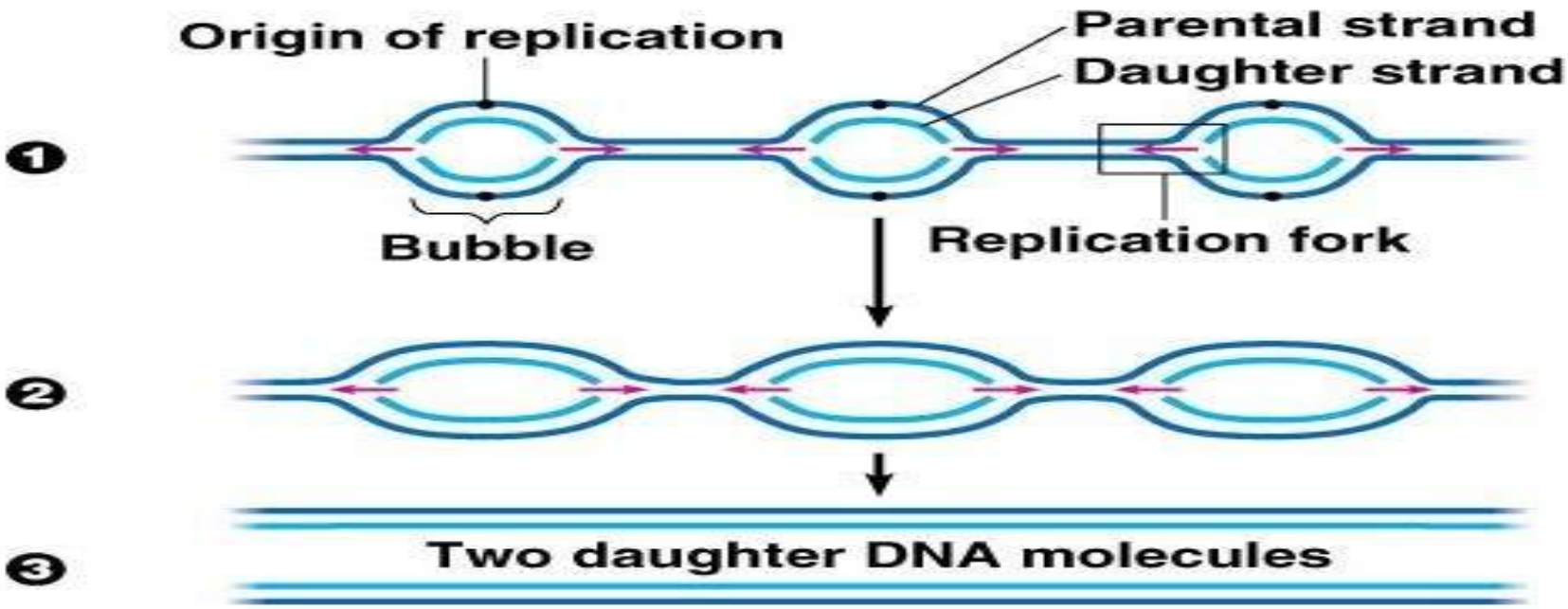
# Replication Fork



# Formation of Replication Bubbles

- Replication occurs in both directions along the length of DNA and both strands are replicated simultaneously.
- This replication process generates **"replication bubbles"**

# Replication Bubbles




(a) In eukaryotes, DNA replication begins at many sites along the giant DNA molecule of each chromosome.

# The DNA Polymerase Complex

- A number of different DNA polymerase molecules engage in DNA replication. These share three important properties: (1) **chain elongation,** (2) **Processivity,** and (3) **proofreading.**
- Chain elongation accounts for the rate (in nucleotides per second) at which polymerization occurs.
- Processivity is an expression of the number of nucleotides added to the nascent chain before the polymerase disengages from the
- template.

  The proofreading function identifies copying errors and corrects them

# DNA Polymerase Complex

- In *E coli,* polymerase III (pol III) functions at the replication fork. Of all polymerases, it catalyzes the highest rate of chain elongation and is the most processive.
- Polymerase II (pol II) is mostly involved in proofreading and DNA repair.
- Polymerase I (pol I) completes chain synthesis between Okazaki fragments on the lagging strand.

# Differences between DNA Polymerase I, II and III

| | DNA pol I | DNA pol II | DNA pol III |
|---|---|---|---|
| Polymerization Rate | Low | Low | High |
| Processivity | Low | Low | High |
| Proof reading | 3'-5' and 5'-3' Exonuclease activities | 3'-5' Exonuclease activity | 3'-5' Exonuclease activity |
| Primer removal | Best | Nil | Nil |
| Strand synthesis | Lagging strand | No role | Both strands |
| DNA repair | Active | Active | No role |

# Eukaryotic DNA polymerases

- Eukaryotic cells have counterparts for each of these enzymes plus some additional ones. A comparison is shown in Table-

| *E coli* | Mammalian | Function |
|---|---|---|
| I | Alpha | Gap filling and synthesis of lagging strand |
| II | Epsilon | DNA proofreading and repair |
| | βeta | DNA repair |
| | Gamma | Mitochondrial DNA synthesis |
| III | delta | Processive , Leading strand synthesis |

# Initiation & Elongation of DNA Synthesis

- Primer-The priming process involves the nucleophilic attack by the 3'-hydroxyl group of the RNA primer on the phosphate of the first entering deoxynucleoside triphosphate with the splitting off of pyrophosphate.
- Mammalian DNA polymerase Alpha is mainly responsible for the synthesis of primer.

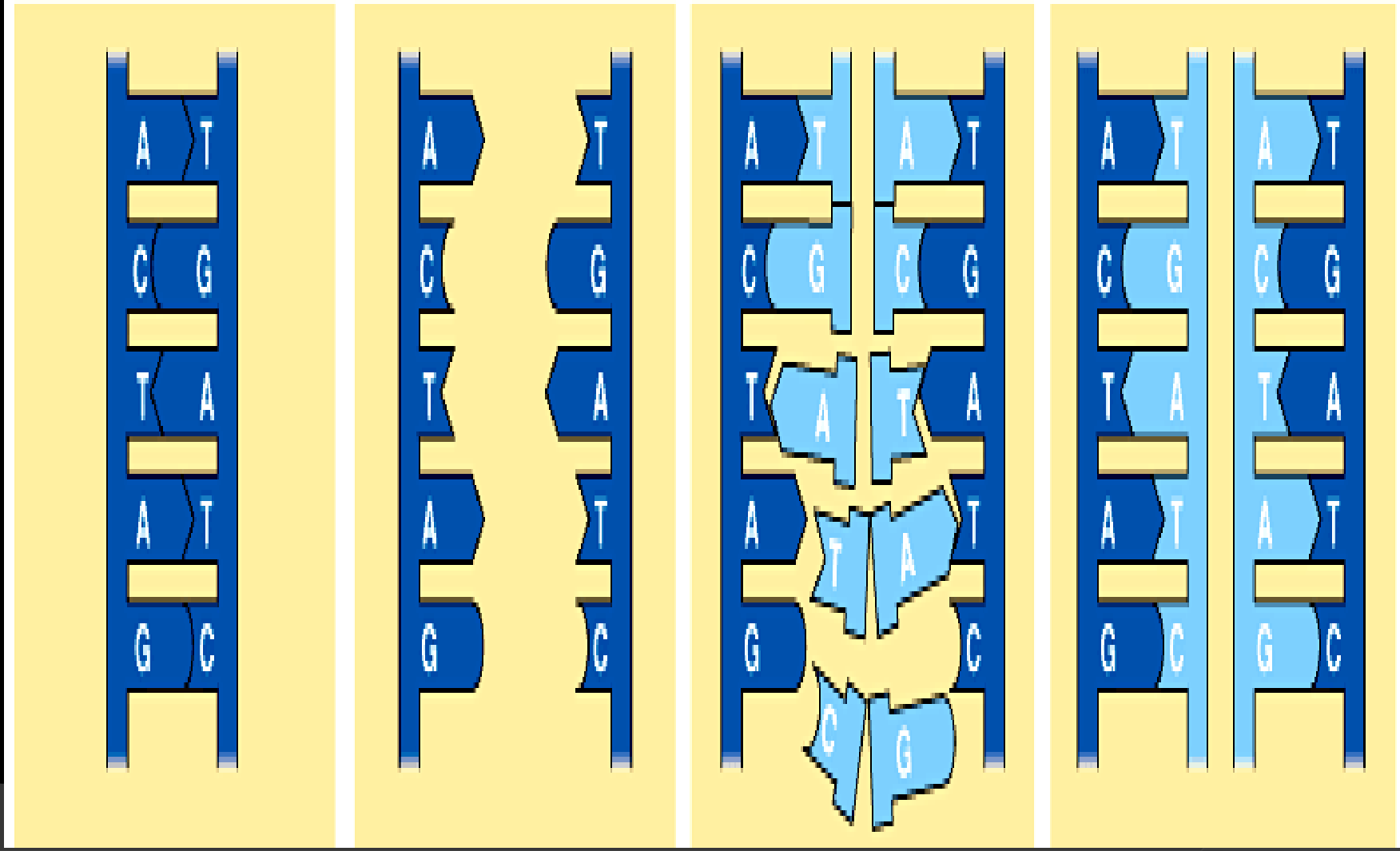
# Initiation & Elongation of DNA Synthesis

- Selection of the proper deoxyribonucleotide whose terminal 3'-hydroxyl group is to be attacked is dependent upon **proper base pairing with the other strand** of the DNA molecule according to the rules proposed originally by Watson and Crick

# Initiation & Elongation of DNA Synthesis

- When an adenine deoxyribonucleoside monophosphoryl moiety is in the template position, a thymidine triphosphate will enter and its phosphate will be attacked by the 3'-hydroxyl group of the deoxyribonucleoside monophosphoryl most recently added to the polymer.
- By this stepwise process, the template dictates which deoxyribonucleoside triphosphate is complementary and by hydrogen bonding holds it in place while the 3'-hydroxyl group of the growing strand attacks and incorporates the new nucleotide into the polymer.

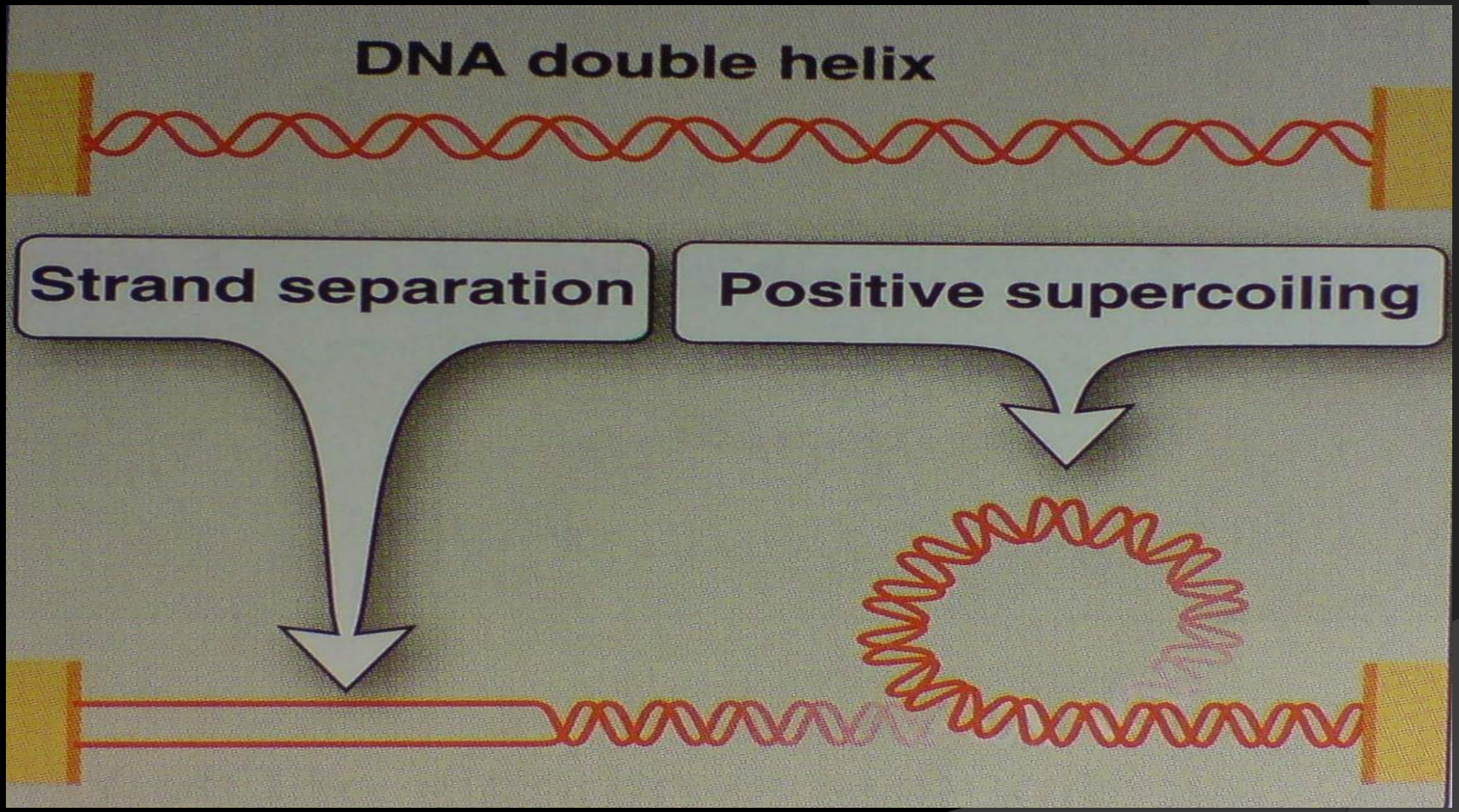
# Base pairing in DNA Replication



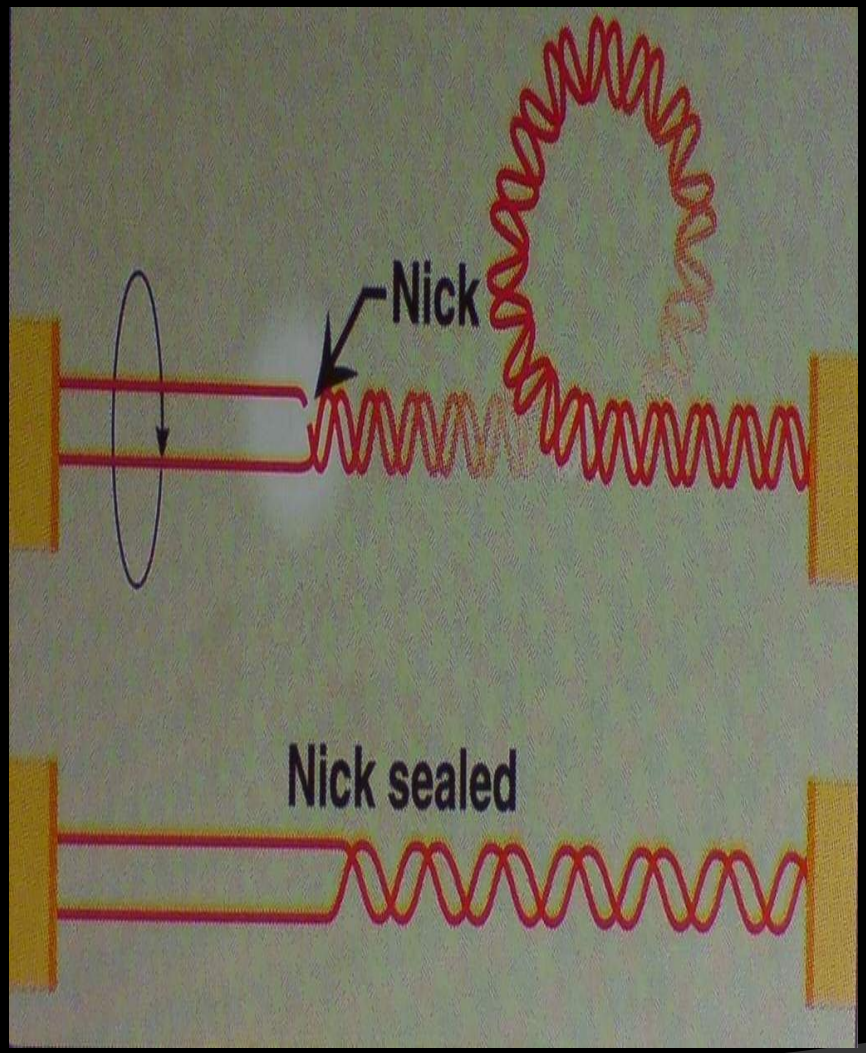
# DNA Topo isomerases

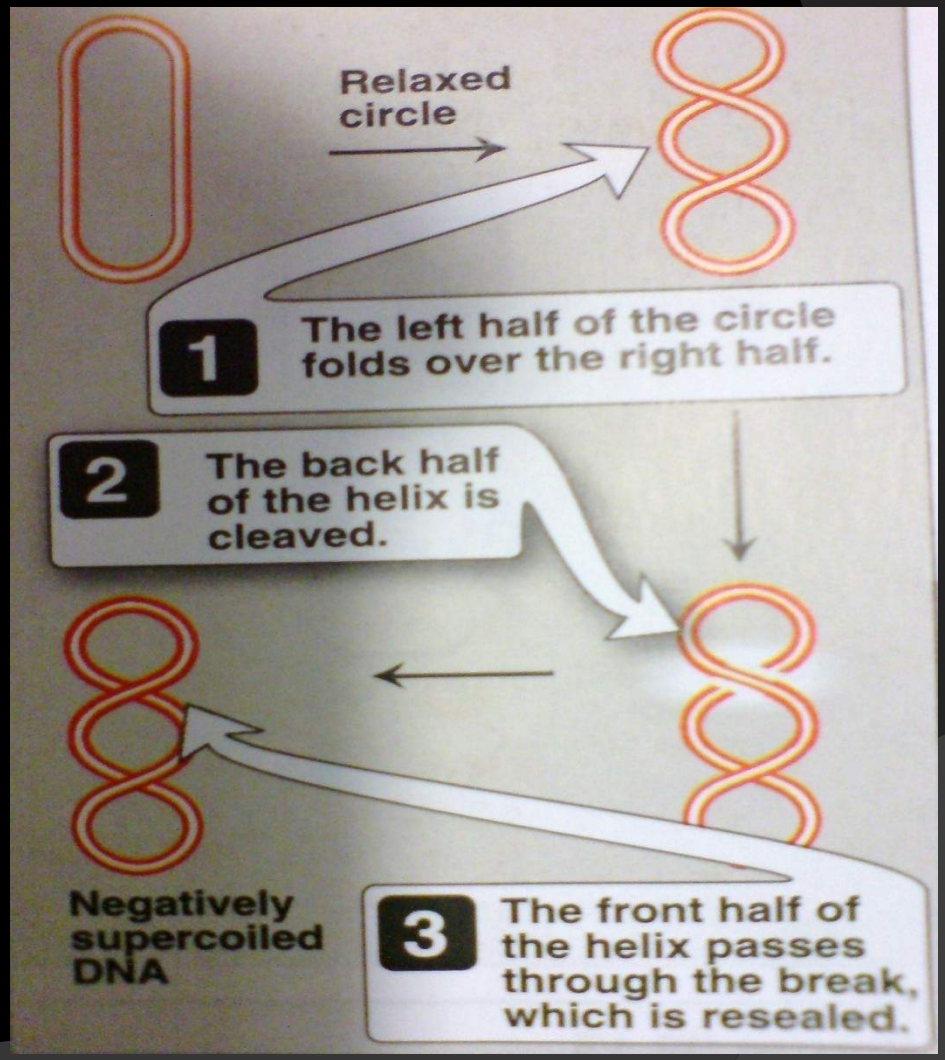
- Relief of super coils is dome by Topo isomerases
- Two types:
- **Topoisomerases I** : acts by making a transient single cut in the backbone of the DNA, enabling the strands to swivel around each other to remove the build-up of twists
- **Topoisomerase II** (DNA Gyrase) acts by introducing double standed breaks enabling one double-stranded DNA to pass through another, thereby removing knots and entanglements that can form within and between DNA molecules.

# Formation of super coils



# Mechanism of action of Topo isomerases(Type-I and Type-II)



# Primer removal and Nick sealing

- Primers are removed by DNA polymerase I by replacing ribonucleotides with deoxy Ribonucleotides
- Nicks are sealed by DNA
- ligase

Multiple primers on the Lagging strand while single primer on the leading strand.

# Proof-reading and Editing

- 1000 bases/second = lots of typos!
- DNA polymerase I
  - proofreads & corrects typos
  - repairs mismatched bases
    - repairs damage throughout life
  - removes abnormal bases
  - reduces error rate from 1 in 10,000 to 1 in 100 million bases

# Termination of replication

- **In prokaryotes**:

  DNA replication terminates when replication forks reach specific "termination sites".

- the two replication forks meet each other on the opposite end of the parental circular DNA .

# Termination of replication

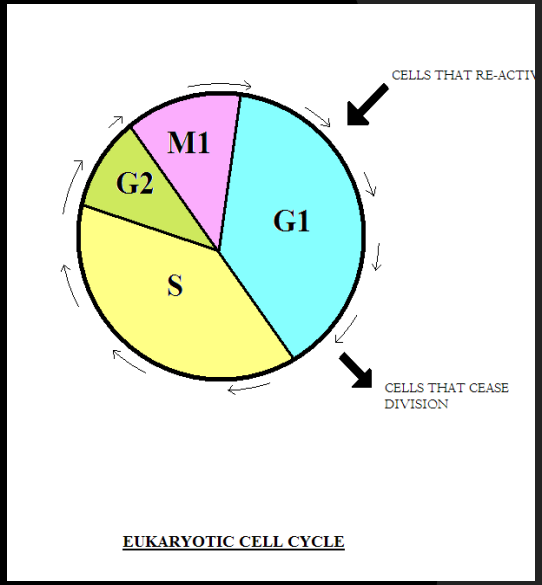
- This process is completed in about 30 minutes, a replication rate of 3 x $10^5$ bp/min in prokaryotes
- The entire mammalian genome replicates in approximately 9 hours, the average period required for formation of a tetraploid genome from a diploid genome in a replicating cell.

# Reconstitution of Chromatin Structure

- chromatin structure must be re-formed after replication. Newly replicated DNA is rapidly assembled into nucleosomes, and the preexisting and newly assembled histone octamers are randomly distributed to each arm of the replication fork.

# DNA Synthesis and the Cell Cycle

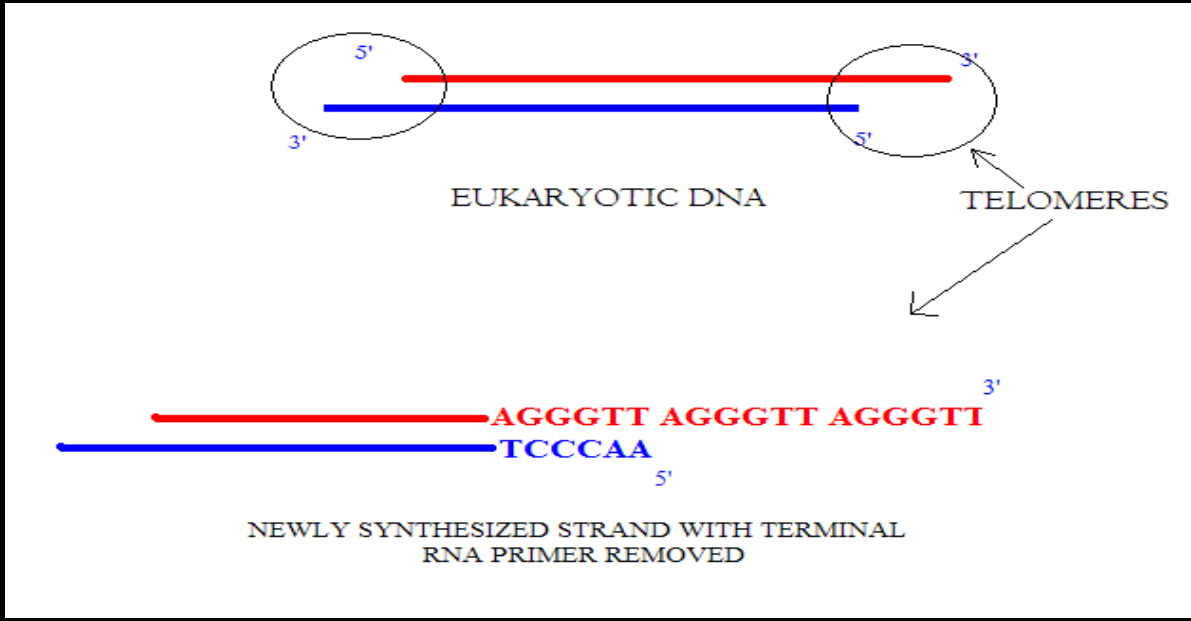
- In animal cells, including human cells, the replication of the DNA genome occurs only during the synthetic or S phase.
- This is usually temporally separated from the mitotic phase by non synthetic periods referred to as gap 1 (G1) and gap 2 (G2), occurring before and after the S phase, respectively
- The cell prepares for DNA synthesis in G1 and for mitosis in G2.



# Telomeres

- In eukaryotic replication, following removal of RNA Primer from the 5’end of lagging strand; a gap is left.
- This gap exposes DNA strand to attack of 5’ exonucleases.
- This problem is overcome by Telomerase.

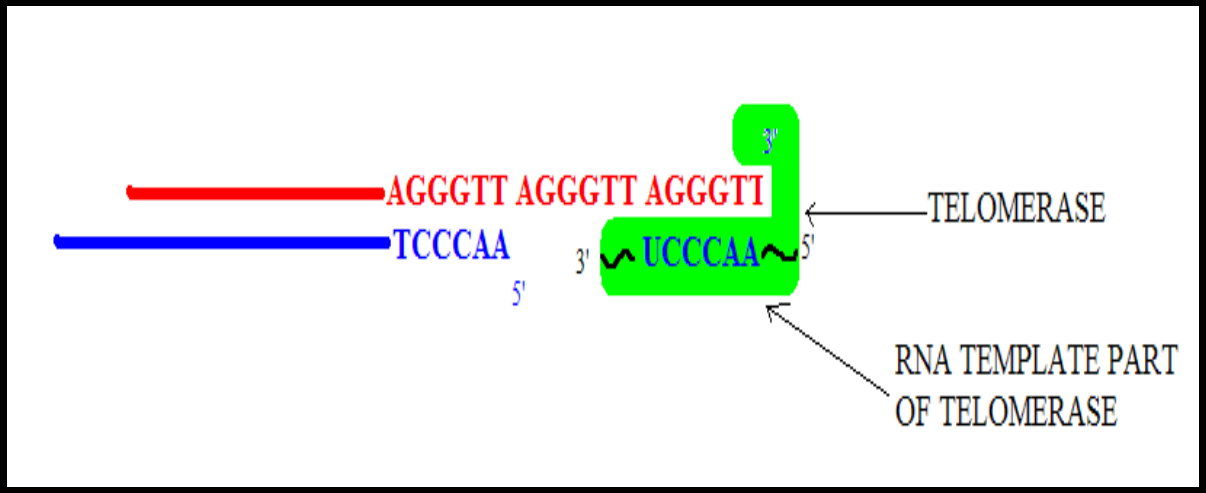
# Telomeres




**Repeating, non-coding sequences at the end of chromosomes = protective cap**

- **limit to ~50 cell divisions**

# Mechanism of action of Telomerase



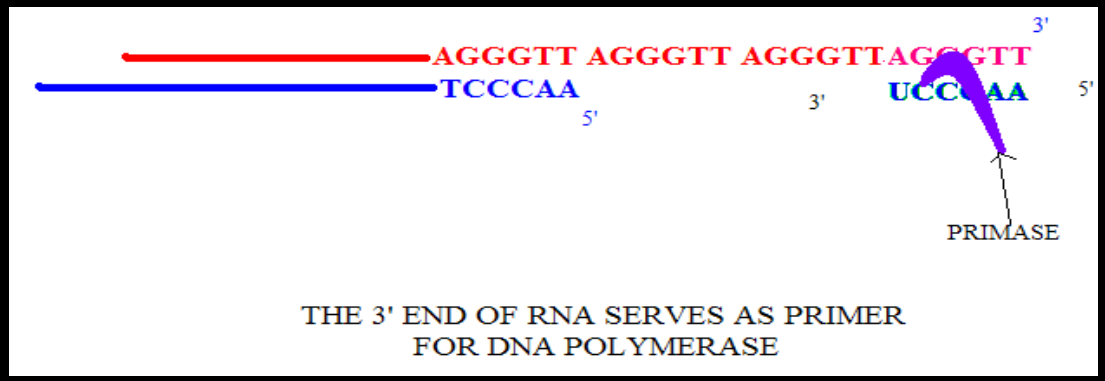

The enzyme synthesizes (TTAGGG)n repeats on to the Telomere sequences, using an internal RNA template.

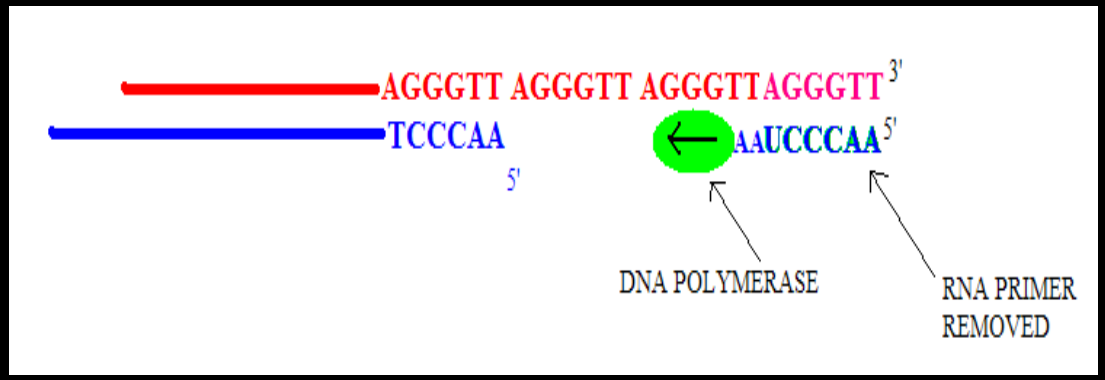
# Mechanism of action of Telomerase(Contd.)




Telomerase acts like a reverse Transcriptase. It recognizes 3' end of telomere, based on the RNA component, a small DNA strand is synthesized

# Mechanism of action of Telomerase(Contd.)




## Telomerase

- different level of activity in different cells
  - High in stem cells & cancers
  - Activity lost in old age
  - Potential target for newer anticancer drugs

# Inhibitors of DNA replication

- Bacterial DNA Gyrase(Type II Topoisomerase)- Inhibited by Novobiocin and Nalidixic acid.
- Ciprofloxacin interferes with DNA breakage and rejoining process
- Mammalian topoisomerases – inhibited by Etoposide and Adriamycin, used as anticancer drugs.
- Nucleoside analogues also inhibit replication and are used as anticancer drugs.

# Summary of Replication

- Unwinding- forms replication fork
- Primase- Synthesizes RNA primer
- Continuous synthesis -Leading strand
- Discontinuous synthesis – Lagging strand (Okazaki fragments)
- Synthesis 5’-3’ direction
- Primers removed, nick sealed
- Proof reading by DNA polymerases
- Organized in to chromatin structure